ICT411 Lab

Course: Cloud Computing
Topic: Introduction to Cloud Computing – Characteristics, Service Models, Deployment Models, Cloud Architecture
Assignment Type: Group work
Due Date: 16/09/2026

Objective

To provide students with direct experience interacting with real cloud platforms, understanding service models, and exploring cloud features through practical exercises.

Lab Instructions

Part 1 – Account Creation & Setup

1. **Choose one** cloud platform from the list below:
    - **Amazon Web Services (AWS)** – https://aws.amazon.com/free
    - **Microsoft Azure** – https://azure.microsoft.com/free
    - **Google Cloud Platform (GCP)** – https://cloud.google.com/free
2. Create a **free-tier account** using your student email.
3. Take **two screenshots**:
    - Screenshot 1: The dashboard/homepage after logging in.
    - Screenshot 2: The "Billing" or "Free Tier Usage" section showing that your account is active.

Part 2 – Service Model Interaction

Perform the following tasks **on your chosen platform**:

A. IaaS – Infrastructure as a Service

- Deploy a **virtual machine** (VM) with any operating system.
- Take a screenshot showing the VM running, with its name, public IP, and status.

C. SaaS – Software as a Service

- Identify a SaaS product within your chosen cloud provider (e.g., Google Docs in GCP, Power BI in Azure, Amazon Chime in AWS).
- Use it briefly (e.g., create a file, dashboard, or meeting) and take a screenshot showing the activity.

Part 3 – Cloud Architecture Mapping

1. Draw a **custom diagram** showing how the services you interacted with (VM, PaaS app, SaaS tool) fit into a cloud architecture.
2. Indicate **front-end**, **back-end**, and **network** elements.
3. Provide a **150-word explanation** describing how each component interacts.

Part 4 – Cost Monitoring & Forecasting

- Navigate to the Cost Management / Billing section of your platform.
- Generate a cost forecast report for the current month.
- Set a monthly budget alert for $1.00 (or the lowest possible amount) to get familiar with the alerting mechanism.
- Take a screenshot of the cost forecast or the budget alert configuration page.
- Write 100 words on why cost monitoring is critical, based on what you saw in the billing dashboard.

Part 5 – Reflection on Advantages and Challenges

Based on your own hands-on experience, write at least five advantages and five challenges you personally encountered while using the cloud platform.

- These must be specific to your experience (e.g., “The AWS EC2 setup wizard was confusing because…” rather than generic answers like “Cloud is scalable”).

Submission Requirements

- Provide a report with screenshots your work. All screenshots must be **unique** (no generic images or cropped stock photos).
- Diagrams can be drawn by hand and scanned, or created digitally (draw.io, Lucidchart, etc.).
- References are not required unless citing external tutorials.